Xingyu Pan

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Address: Beijing Normal University, Zhuhai, Guangdong Province, China

Education Background

Beijing Normal University, Zhuhai, China Sep 2025 - Present

Ph.D. Student in Psychology (Cognitive Neuroscience)

Supervisor: Dr. Jing Huang

Research Focus: Investigating cognitive and neural mechanisms underlying action–cognition interactions, focusing on visuomotor contributions to visual working memory, and neural dynamics of perceptual learning and transfer.

Beijing Sport University, Beijing, China Sep 2022 - Jun 2025

M.Sc. in Psychology (Cognitive Neuroscience)

Supervisor: Prof. Yu Zhang, Associate Prof. Jie Li

GPA: 3.6/4.0

Research Focus: Investigating attentional allocation during cooperation and individual visual tracking through fNIRS hyperscanning and eye-tracking experiments.

China Agricultural University, Beijing, China Sep 2017 - Jun 2021

B.Sc. in Management

Research Experience

Dr. Jing Huang's Research Group, Beijing Normal University, China Sep 2025 - Present

Visuomotor Integration Moderates the Effect of Physical Activity on Visual Working Memory

- Developed a research project investigating the role of visuomotor integration in the relationship between physical activity and visual working memory.
- Conducted data analysis and robust moderation modeling, identifying visuomotor precision as a boundary condition of the physical activity-WM relationship.
- Wrote and submitted the first-author manuscript.

TMS–EEG Investigation of Cortical Contributions to Visual Working Memory

- Contributed to research planning and experimental setup for an online TMS study targeting attention- and motor-related cortical regions during a retro-cue WM task.
- Participated in preliminary behavioral investigations examining differences in WM performance across stimulation conditions.
- Currently conducting preliminary analyses of concurrent TMS‑EEG data to characterize WM-related neural dynamics under different cue conditions.

Neural Mechanisms of Perceptual Learning and Transfer

- Applied tensor decomposition to EEG data to identify neural activity patterns associated with perceptual learning and transfer.
- Identified distinct training-related alpha-band components associated with different stimulus formats.

- Wrote the first-author manuscript and presented the findings in an oral presentation at the Joint ESCoP–CoPM Meeting 2026.

Prof. Yu Zhang's Research Group, Beijing Sport University, China Sep 2022 - Jun 2025

Neural Mechanisms of Attentional Allocation in Cooperative Tasks Among Athletes *Master's Thesis*

- Developed a cooperative multiple object tracking paradigm to investigate attentional allocation during cooperative tasks.
- Conducted fNIRS hyperscanning analyses to examine functional connectivity and inter-brain synchrony during cooperation.

Goal-Directed Attentional Allocation in Dynamic Visual Tracking: Evidence from Eye Tracking

- Contributed to the design of eye-tracking experiments investigating goal-directed visual attention.
- Analyzed gaze patterns, identifying shifts from central fixation to target-directed fixations as attentional demands increased.

Publications

Manuscripts Submitted

Pan, X., Fan, T., Wang, F., Zhou, T., Song, Y., & Huang, J. Format-Specific Processing Demands Determine Transfer of Perceptual Learning. *Manuscript submitted for publication.*

Pan, X., Liu, L., Long, Y., Li, X., Li, Z., Fan, T., & Huang, J. Visuomotor Integration Moderates the Effect of Physical Activity on Visual Working Memory in Young Adults. *Manuscript submitted for publication.*

Conference Papers

Pan, X., Fan, T., Wang, F., Zhou, T., Song, Y., & Huang, J. (2026). From Sensory Tuning to Representational Convergence: Multidimensional EEG Dynamics of Visual Perceptual Learning. *Joint ESCoP–CoPM Meeting 2026*. Oral presentation.

Pan, X., Li, J., & Zhang, Y. (2023). The Influence of Depth on Visual Attention in 3D Space. *25th National Academic Congress of Psychology*, China. Poster presentation.

Honors & Awards

- Scholarship, Beijing Normal University *2025, 2026*
- Graduate Research Excellence and Innovation Program, Beijing Sport University. *Project Lead*; *university-funded graduate research project*. *2024*

Research & Technical Skills

Data Acquisition: EEG, Eye Tracking, fNIRS Hyperscanning

Data Analysis: ERP, Time–Frequency Analysis, Tensor Decomposition, Eye-Movement Analysis, Functional Connectivity, Inter-Brain Synchrony

Programming & Software: MATLAB, R